

PROJECT REPORT: BANK CUSTOMER ANALYSIS

Strategic Prediction of Product Subscription using Decision Trees

1. Executive Summary

This project successfully implemented a Decision Tree Classifier to predict whether clients would subscribe to a bank term deposit. By prioritizing **Recall** over simple accuracy, we developed a model that identifies the vast majority of potential customers, directly supporting the bank's revenue goals.

2. Data Pipeline (Step-by-Step)

I. Data Cleaning

- **Unknown Handling:** Categorical "unknowns" were retained as a separate class to capture latent patterns.
- **Temporal Correction:** Value 999 in 'pdays' was mapped to -1 to represent new leads correctly.

II. Feature Engineering

All categorical features (Job, Marital, Education) were transformed using **One-Hot Encoding**. The target variable 'y' was mapped to binary integers (0 and 1).

III. Balancing the Scale

Because the dataset is imbalanced (more 'No' than 'Yes'), we used the `class_weight='balanced'` parameter. This is the "secret sauce" that allows the model to find subscribers effectively.

3. Final Model Performance

80.73%	79.04%	0.84	0.75
Training Accuracy	Testing Accuracy	Recall (Success)	Precision (Success)

Note: While the accuracy is ~80%, the 0.84 Recall means the model identifies 84% of all actual subscribers. This is a superior business result compared to a 90% accurate model that fails to find buyers.

4. Classification Insights

Category	Precision	Recall	F1-Score
No (Non-Subscribers)	0.83	0.75	0.79
Yes (Subscribers)	0.75	0.84	0.79

5. Conclusion & Strategy

The Decision Tree reveals that **call duration** and **previous success** are critical. The bank should focus its sales force on the segments identified by the 'Yes' leaf nodes in this tree, as they represent an 84% capture rate of the available market.